

Colin Buchheit

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 github |  linkedin

 colinbuchheit.com

EDUCATION

Bachelor of Science - Information Technology

May 2025

University of Missouri - Columbia

Columbia, MO

- Cum Laude, High Dean's List
- GPA: 3.78, Major GPA: 3.96
- Clubs: Mizzou Computing Association, The Mizzou Space Program

Associate Degree

Aug 2023

St. Louis Community College

St. Louis, MO

- Member of Phi Theta Kappa - Honors Fraternity

SKILLS

Programming Languages	TypeScript Python Java C# JavaScript Bash
Web Development	React Node.js Angular React-Redux FastAPI HTML/CSS
Cloud & DevOps	Docker Azure DevOps GitHub Actions CI/CD VMware VirtualBox
Databases & AI/ML	MongoDB PostgreSQL MySQL Redis Azure AI Vision Azure Speech

EXPERIENCE

MX Holdings

May 2024 - Jan 2025

Enterprise Software Developer Intern

St. Louis, MO

- Developed a real-time diagnostics dashboard to track and diagnose critical errors, reducing down-time and improving response times.
- Created an internal market pricing interface web application aggregating data from international metal exchanges to enhance price transparency and aid strategic decisions.
- Optimized system performance through full-stack development, enhancing back-end infrastructure and refining front-end UI with Redux and TypeScript.
- Developed AI-powered customer product purchasing application with intent recognition and multi-stream departmental connections, enabling intelligent product availability insights and automated purchasing through conversational interface.

NOTABLE PROJECTS

TigerHacks Hackathon - 1st Place Winner

Fall 2023

Rapid Reels - Short-Form Video Platform

Columbia, MO

- Team of 4 to develop winning project among 180+ participants and 33 submitted projects in top category.
- Designed and built scalable short-form video platform showcasing innovation and cross-functional teamwork under tight time constraints.

Discord AI Assistant - Senior Capstone

Fall 2024 - Deployed

AI Chat Agent

Columbia, MO

- Designed 24/7 AI assistant for academic support via Discord using TypeScript, Python, Discord.js, MongoDB, and Docker.
- Implemented layered architecture (API, application, domain, infrastructure) ensuring scalability and security.
- Integrated syllabus-based training, multi-persona AI interaction, conversation memory, and debugging features.

AI Natural Language Database Query Tool (MaiQuery)

Ongoing

Columbia, MO

Full Stack

- Developing full-stack application enabling natural language database queries using AI agent network with TypeScript, React, Node.js, Python, FastAPI, and CrewAI.
- Implemented support for SQL and NoSQL databases with schema-aware query generation and real-time WebSocket streaming.
- Architected modular microservice infrastructure with Docker containerization for scalable cloud and local deployments.

CERTIFICATIONS

- **Microsoft AI & ML Engineering Certificate** (Coursera) - May 2025
- **Cybersecurity Certificate** (University of Missouri) - May 2025
- **Information Systems and Technology Certificate** (University of Missouri) - May 2025